

Torque Ripple Suppression of BLDCM With Optimal Duty Cycle and Switch State by FCS-MPC

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ABSTRACT The traditional six-step commutation control method in brushless DC motors (BLDCM) often results in significant torque ripple, which limits its application in high-precision fields. This paper proposes a scheme that combines finite control set model predictive control (FCS-MPC) to suppress commutation torque ripple. The scheme utilizes an optimal duty cycle and optimal switch compensation state. By analyzing the mathematical model of BLDC motors, the causes of commutation torque ripple were identified. To maintain the slope of the non-commutation current unchanged, a discrete model was used to predict the non-commutation current at the next moment. The predicted current was then used to calculate the optimal duty cycle. To address the shortcomings of PWM regulation performance at high speeds, a switch insertion compensation strategy was proposed. The strategy selects the optimal switch insertion compensation state based on a cost function to offset the changes in non-commutation current. Furthermore, feedback compensation was incorporated to enhance robustness in the event of parameter mismatch. The effectiveness of this method was demonstrated through experimental and simulation results.

INDEX TERMS BLDCM, model predictive control, torque ripple, cost function.

I. INTRODUCTION

The brushless DC motor (BLDCM) is a popular type of motor in contemporary industry. It finds wide applications in various industries such as home appliances, cars, medical equipment, and aerospace due to its dependability, high power density, and high efficiency [1]. However, in high-precision applications, the torque ripple problem of BLDCM has always been a significant factor that limits their performance. Excessive torque pulsation can negatively impact the stability of load output and speed, as well as increase motor energy consumption.

There are multiple factors that can cause unstable motor operation. The stator current waveforms can be distorted and cause torque ripple when the rotor and stator winding parameters are not compatible. Another factor is the uneven distribution of the air gap magnetic flux density, caused by the asymmetry of the magnetic circuit. Additionally, unbalanced

mechanical loads can cause rotor vibration, leading to torque pulsation. Finally, improper control algorithms can distort the stator current waveform, resulting in torque ripple. The hardware is responsible for the former three torque pulsations, and efforts should be made to minimize them during motor design and production. On the other hand, torque ripple caused by poor control algorithms can be minimized at the software level [2].

Numerous recent studies have focused on torque ripple in BLDCM. In the context of DC voltage modulation, reference [3] suggests an improved combination of a single-ended primary inductor converter and DC regulator that can be utilized in three-level inverters to lessen the torque ripple of BLDCM. Reference [4] proposes an auxiliary boost converter that can amplify the input voltage to quadruple the amplitude of back electromotive force (BEMF), thereby minimizing the torque ripple during commutation. According to reference [5],

a single switch can be used to predict the output voltage and voltage compensation technology can be employed to account for torque at a constant speed, eliminating the need for a DC connection capacitor. To enhance the converter's efficiency, [6] and [7] propose a novel zero-energy thermomuclear component converter and interleaved boost converter respectively. These techniques effectively suppress torque ripple, but require additional topology circuits.

PWM has been widely used to implement the DC side voltage regulation method, aiming to reduce the hardware cost of the DC side. Among the five popular PWM modulation techniques, PWM_ON exhibits the least modulation torque ripple. In reference [8], H_PWM_L_PWM is proposed as an efficient way to minimize torque ripple when motor braking. Another method, PWM_ON_PWM, is proposed in reference [9], and simulation and experimentation have shown that it has lower torque ripple compared to PWM_ON. Implementing this method on hardware circuits is challenging as it requires precise acquisition of the rotor position and accurate duty cycle calculation for PWM modulation. To address this, model predictive control (MPC) is utilized in references [10], [11], [12] for duty cycle determination due to its effectiveness in reducing torque ripple compared to PI control. However, due to the limited number of PWM cycles, their conversion into equivalent voltage during commutation has negligible impact on torque ripple suppression for high-speed motor rotation. To mitigate high-speed torque ripple, three conduction and two conduction switching control modes are discussed in references [13], [14], [15]. Additionally, in reference [16], a twelve-step direct torque control with minimum overlap angle is proposed as a means to successfully reduce torque ripple caused by six-step commutation control.

Some researchers are currently integrating complex control algorithms into the current control loop to reduce torque ripple. A method to reconstruct the ideal current and torque waveform trajectory by using torque and current waveform analysis is proposed in [17]. The influence of various loss factors on the actual output current was analyzed by finite element simulation. References [18] and [19] employed synovial observer and Luenberger observer to enhance the reference current optimization, diminishing the input steady-state error within the current loop. In [20] and [21], the least mean squares adaptive neuron scheme and genetic algorithm are used to robustly optimize PI parameters of the current loop, respectively. However, this method has a large amount of calculation and takes up a lot of memory. For the maximum torque per ampere control, a power predictive control is proposed in [22] and an optimal turn-on and turn-off commutation angle control is proposed in [23]. In reference [24], considering the influence of ferromagnetic loss, a direct torque switching vector compensation control method is adopted.

In this paper, a control strategy based on model prediction of duty cycle and switch state is proposed for the simplest six step commutation control of brushless DC motors. The main characteristics and contributions of this article are as follows:

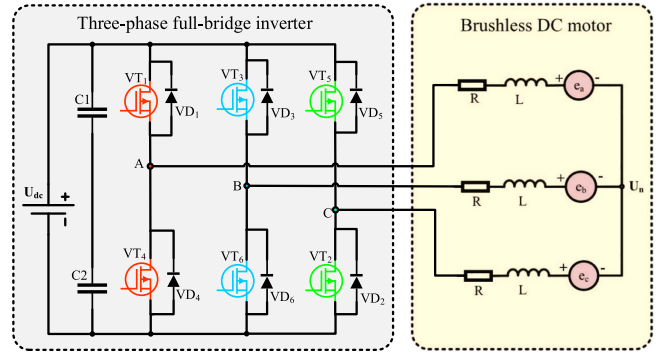


FIGURE 1. Three-phase inverter circuit of BLDCM.

- Adopting PWM_ON_PWM bilateral modulation technology can significantly reduce torque ripple compared with traditional PWM_ON unilateral modulation.
- The traditional PI current loop is replaced by the duty cycle predicted by the model, which reduces the complexity of PI parameter tuning and the steady-state error caused by the change of nonlinear parameters.
- The compensation effect of non-commutation current under different switch conduction modes is analyzed. Based on the cost function, we choose the optimal state for switch insertion to ensure a constant non-commutation current. This method is more effective for torque ripple suppression at high speeds.
- Through feedback correction, the interference of parameter mismatch on model prediction is reduced.

The subsequent sections of this document are structured as follows: In Section II, an analysis is carried out on the underlying factors leading to torque pulsation. The methods for implementing control algorithms are presented in Section III. Furthermore, the simulation findings are shown in Section IV, while the experimental verifications are provided in Section V. Lastly, the overall conclusion of this article can be found in Section VI.

II. SOURCE OF COMMUTATION TORQUE RIPPLE

The analogous circuit for BLDCM in the classical topology of a six-switch inverter is presented in Fig. 1. The switch tube VT₁-VT₆ is used to complete the electronic commutation and inverter output, and the anti-parallel diode VD₁-VD₆ is used for freewheeling.

The mathematical model of BLDCM is described as follows:

$$\begin{bmatrix} U_a \\ U_b \\ U_c \end{bmatrix} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} L & 0 & 0 \\ 0 & L & 0 \\ 0 & 0 & L \end{bmatrix} \cdot \begin{bmatrix} \frac{di_a}{dt} \\ \frac{di_b}{dt} \\ \frac{di_c}{dt} \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} + \begin{bmatrix} U_n \\ U_n \\ U_n \end{bmatrix} \quad (1)$$

The stator terminal voltages of the three phases are represented as U_a , U_b and U_c . The midpoint voltage of the motor is denoted as U_n . Similarly, the stator currents are indicated as i_a , i_b and i_c . The three-phase stator winding back electromotive forces (BEMFs) are known as e_a , e_b and e_c . Winding resistance is represented by R and winding inductance is denoted as L .

The relation between three-phase currents is:

$$i_a + i_b + i_c = 0 \quad (2)$$

The basic formula for electromagnetic torque is:

$$T_e = \frac{e_a i_a + e_b i_b + e_c i_c}{\omega_m} \quad (3)$$

where ω_m represents mechanical angular velocity.

The magnitude of the BEMF is directly proportional to the angular velocity:

$$E = k_e \cdot \omega_m \quad (4)$$

where k_e is the BEMF constant.

For example, when current passes through the freewheeling diode during phase A, which is the non-commutation phase, and the lower switch of phase B is off, the lower switch of phase C is turned on, conduction from A+B- to A+C- occurs. The values of three-phase voltages and BEMFs are as follows:

$$u_a = U_{dc}, u_b = U_{dc}, u_c = 0 \quad (5)$$

$$e_a = E, e_b = -E, e_c = -E \quad (6)$$

Substitute equations (5) and (6) into (1), disregarding the stator winding resistance, the equation is:

$$\begin{cases} \frac{d(i_a)}{dt} = \frac{U_{dc}-4E}{3L} \\ \frac{d(i_b)}{dt} = \frac{U_{dc}+2E}{3L} \\ \frac{d(i_c)}{dt} = \frac{2E-U_{dc}}{3L} \end{cases} \quad (7)$$

Combined with (3) and (7), torque ripple during commutation is:

$$\Delta T_e = \frac{\Delta P_0}{\omega_m} = \frac{2E(U_{dc}-4E)}{3L \cdot \omega_m} \quad (8)$$

From (8), the motor will produce commutation torque ripple when $U_{dc} - 4E \neq 0$.

Substituting (6) into (3), combined with (2):

$$T_e = \frac{E i_a - E i_b - E i_c}{\omega_m} = \frac{2E i_a}{\omega_m} \quad (9)$$

(9) demonstrates that when ω_m at a constant velocity, the electromagnetic torque exhibits a direct correlation with the non-commutation phase current. It is thus advised to maintain a constant non-commutation phase current in order to minimize commutation torque ripple. The inductor assumes a critical role in preventing sudden fluctuations in the current throughout the commutation procedure. The varying rates of current alteration during commutation are depicted in Fig. 2.

When the BLDCM is operating at a low velocity, the voltage on the DC side exceeds four times the magnitude

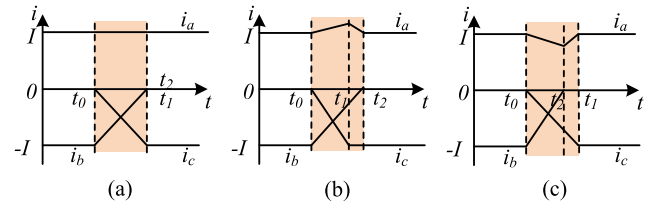


FIGURE 2. Current commutation behaviors during different speed. (a) $U_{dc} = 4E$, (b) $U_{dc} > 4E$, (c) $U_{dc} < 4E$.

of the BEMF. Furthermore, in phase A, the positive rate of change in phase current can be seen in Fig. 2(b). On the contrary, when the BLDCM is running at a high speed, the DC side voltage is lower than four times the value of the BEMF, and the negative rate of change of the phase current in phase A during the non-commutation phase is shown in Fig. 2(c). In both cases, torque pulsations occur. To effectively mitigate torque pulsation, it is crucial to adjust the current rate of change of incoming and outgoing phases to keep the non-commutation phase current constant, as demonstrated in Fig. 2(a).

III. SUPPRESSION STRATEGY OF COMMUTATION TORQUE RIPPLE

FCS-MPC is a control strategy developed on the basis of predictive control, and the discrete algorithm that determines the input sequence by predicting the future state of the system. This study uses FCS-MPC to predict the current of non-commutation phase at the subsequent instant. Following feedback correction, the best state among the three conduction states is then chosen as the switch state at the subsequent instant through a cost function. The control system is depicted as a block diagram in Fig. 3.

A. ESTABLISHING THE PREDICTIVE MODEL

In the case of transitioning from commutation A+B to A+C, the (1) can be transposed into the following using the BLDC mathematical model:

$$u_b - u_a = R(i_b - i_a) + L \left(\frac{di_b}{dt} - \frac{di_a}{dt} \right) + (e_b - e_a) \quad (10)$$

$$u_c - u_a = R(i_c - i_a) + L \left(\frac{di_c}{dt} - \frac{di_a}{dt} \right) + (e_c - e_a) \quad (11)$$

Substituting (2) into (10) get:

$$u_b - u_a = R(-i_c + 2i_a) + L \left(-\frac{di_c}{dt} - 2\frac{di_a}{dt} \right) + (e_b - e_a) \quad (12)$$

Arranging equations (11) and (12) into:

$$u_b + u_c - 2u_a = -3Ri_a - 3L\frac{di_a}{dt} + (e_b + e_c - 2e_a) \quad (13)$$

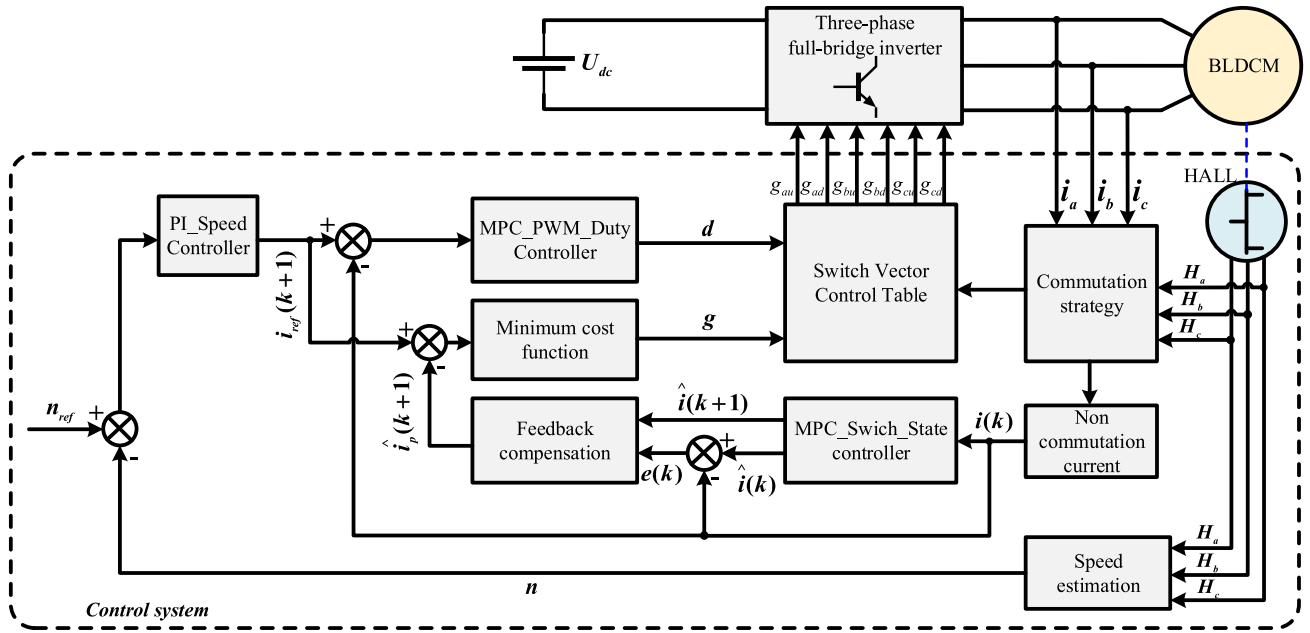


FIGURE 3. System control block diagram.

According to the forward Euler's formula:

$$\frac{d(i_a)}{dt} = \frac{i_a(k+1) - i_a(k)}{T_c} \quad (14)$$

where T_c is the sampling time.

Substituting (14) into (13), The discretized model for predicting current can be represented as:

$$i_a(k+1) = \frac{-(u_b(k) + u_c(k) - 2u_a(k))T_c}{3L} + \frac{(3L - 3RT_c)i_a(k) + (e_b(k) + e_c(k) - 2e_a(k))T_c}{3L} \quad (15)$$

There are six commutation processes during the operation of the motor. After sorting and analyzing the six commutation processes, the predicted current model is:

$$\hat{i}_x(k+1) = \frac{-(u_y(k) + u_z(k) - 2u_x(k))T_c}{3L} + \frac{(3L - 3RT_c)i_x(k) + (e_y(k) + e_z(k) - 2e_x(k))T_c}{3L} \quad (16)$$

The non-commutation phase is represented by x , the outgoing-phase by y , and the incoming-phase by z . According to the aforementioned formula, the three-phase winding voltage value is related to the switching state of the switch tube, and the predicted current of the non-commutation phase at subsequent instant is also related to the output current value at k moment.

B. CALCULATION OF OPTIMAL PWM DUTY CYCLE

PWM technology is widely used for voltage regulation of DC motors, and this article adopts PWM_ON_PWM modulation method. This method is a bilateral modulation technique specifically used for six-step commutation control of BLDCM, which has a smaller torque ripple compared to other PWM modulation methods in [9].

Here is the same example of the commutation phase as above. If $U_{dc} > 4E$, the ideal voltages in three-phase are:

$$u_a = U_{dc}, u_b = U_{dc}, u_c = (1 - d_1) * U_{dc} \quad (17)$$

where d_1 corresponds to the duty cycle of the incoming phase. Substituting (17) and (6) into (1):

$$\begin{cases} U_a = R \cdot i_a + L \frac{di_a}{dt} + e_a + U_n = U_{dc} \\ U_b = R \cdot i_b + L \frac{di_b}{dt} + e_b + U_n = U_{dc} \\ U_c = R \cdot i_c + L \frac{di_c}{dt} + e_c + U_n = (1 - d_1)U_{dc} \end{cases} \quad (18)$$

Combining (2), the relationship between non-commutation current and duty cycle d_1 is obtained as:

$$U_{ac} = 3Ri_a + 3L \frac{di_a}{dt} + 4E = d_1 * U_{dc} \quad (19)$$

According to (14), the universal duty cycle d_1 is obtained as:

$$d_1 = \left\{ 3Ri_x(k) + \frac{3L}{T_c} [i_x(k+1) - i_x(k)] + 4E \right\} / U_{dc} \quad (20)$$

where i_x is defined as the current of the non-commutation phase, $i_x(k+1)$ is equal to the reference current $i_{ref}(k+1)$ at the next moment.

When $U_{dc} < 4E$, the ideal voltages in three-phase are:

$$u_a = U_{dc}, u_b = 0, u_c = (1 - d_2) * U_{dc} \quad (21)$$

where d_2 denotes the duty cycle of the outgoing phase.

Substituting (21) and (6) into (1):

$$\begin{cases} U_a = R \cdot i_a + L \frac{di_a}{dt} + e_a + U_n = U_{dc} \\ U_b = R \cdot i_b + L \frac{di_b}{dt} + e_b + U_n = 0 \\ U_c = R \cdot i_c + L \frac{di_c}{dt} + e_c + U_n = (1 - d_2)U_{dc} \end{cases} \quad (22)$$

Combining (2), the relationship between non-commutation current and duty cycle d_2 is obtained as:

$$U_{ac} = 3Ri_a + 3L \frac{di_a}{dt} + 4E - U_{dc} = d_2 * U_{dc} \quad (23)$$

According to (14), the universal duty cycle d_2 is obtained as:

$$d_2 = \left\{ 3Ri_x(k) + \frac{3L}{T_c} [i_x(k+1) - i_x(k)] + 4E - U_{dc} \right\} / U_{dc} \quad (24)$$

The parameters of the BLDCM mathematical model can be employed for duty cycle computation, as illustrated in the aforementioned example. This technique can substitute the traditional current loop of PI and reduce the requirement for parameter adjustment. The expressed optimal duty cycle for bilateral modulation can be described as follows:

$$d = \begin{cases} d_1, & \text{if } U_{dc} \leq 4E \\ d_2, & \text{if } U_{dc} > 4E \end{cases} \quad (25)$$

Adopting PWM_ON_PWM, the duty cycle for both sides of the modulation is d . To prevent over modulation, the d limit is between 0 and 1.

C. SWITCH STATUS COMPENSATION

In the process of transitioning from A+B- to A+C-, the conduction switch is divided into three modes: A+ for single-phase conduction, A+C- for two-phase conduction, and A+B-C for three-phase conduction. The selection of one of these conduction modes during the commutation phase aims to minimize torque ripple. However, it is important to highlight that the compensation provided by the inserted switch state is only effective under constant speed and load conditions.

At the moment of commutation, the switch state must first switch to two-phase conduction A+C- based on the traditional paired conduction state, as shown in Fig. 4.

If this example remains the same as the one from the previous commutation stage, the voltages and BEMFs are same as (5) and (6). Substituting them into (1), the current change rate in phase A is:

$$\frac{di_a}{dt} = \frac{U_{dc} - 4E}{3L} \quad (26)$$

According to (26), in non-commutation, if U_{dc} exceeds $4E$, the current rate becomes positive, suggesting the BLDCM is functioning at a low speed. Conversely, if U_{dc} is below $4E$, the current rate of phase A turns negative, indicating a high-speed operation of the BLDCM.

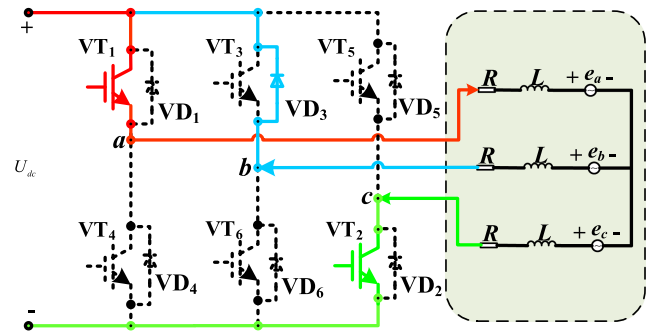


FIGURE 4. Two phase winding conduction of A+C-.

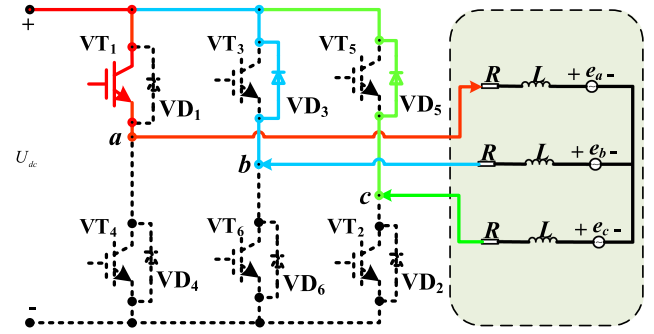


FIGURE 5. Single phase winding conduction of A+.

When the single-phase conduction A+ is switched from the two-phase conduction A+C-, as demonstrated in Fig. 5, the conduction of VT₁ in phase A occurs, while phases B and C freewheel through the diodes of the upper bridge arm. The voltages in three-phase are:

$$u_a = U_{dc}, u_b = U_{dc}, u_c = U_{dc} \quad (27)$$

Substituting (27) and (6) into (1), the current of phase A at the following rate:

$$\frac{di_a}{dt} = \frac{-4E}{3L} \quad (28)$$

At this time, the current rate of change is negative. It can suppress the increase in the current slope during normal conduction.

When transitioning from the A+C- two-phase conduction to the A+B-C- three-phase conduction, as depicted in Fig. 6, the resulting voltages of three-phase are:

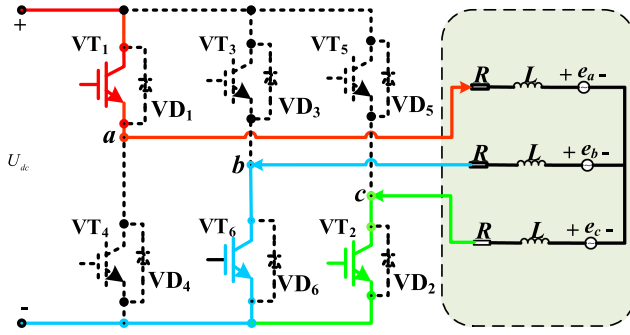
$$u_a = U_{dc}, u_b = 0, u_c = 0 \quad (29)$$

Substituting (29) and (6) into (1), the current of phase A at the following rate:

$$\frac{di_a}{dt} = \frac{2U_{dc} - 4E}{3L} \quad (30)$$

TABLE 1. MPC Switch Vector Control Table

HALL	Commutation	Non-commutation phase	Phase BEMFs	Phase voltages	Minimum cost function	Switch status
101→100	AB→AC	A	$e_a = E, e_b = -E, e_c = -E$	$U_a = U_{dc}, U_b = U_{dc}, U_c = U_{dc}$	g_1	A+
				$U_a = U_{dc}, U_b = U_{dc}, U_c = 0$	g_2	A+C-
				$U_a = U_{dc}, U_b = 0, U_c = 0$	g_3	A+B-C-
				$U_a = 0, U_b = 0, U_c = 0$	g_1	C-
100→110	AC→BC	C	$e_a = E, e_b = E, e_c = -E$	$U_a = 0, U_b = U_{dc}, U_c = 0$	g_2	B+C-
				$U_a = U_{dc}, U_b = U_{dc}, U_c = 0$	g_3	A+B+C-
				$U_a = U_{dc}, U_b = U_{dc}, U_c = U_{dc}$	g_1	B+
				$U_a = 0, U_b = U_{dc}, U_c = U_{dc}$	g_2	B+A-
110→010	BC→BA	B	$e_a = -E, e_b = E, e_c = -E$	$U_a = 0, U_b = U_{dc}, U_c = 0$	g_3	B+C-A-
				$U_a = 0, U_b = 0, U_c = 0$	g_1	A-
				$U_a = 0, U_b = 0, U_c = U_{dc}$	g_2	C+A-
				$U_a = 0, U_b = U_{dc}, U_c = U_{dc}$	g_3	B+C+A-
010→011	BA→CA	A	$e_a = -E, e_b = E, e_c = E$	$U_a = U_{dc}, U_b = U_{dc}, U_c = U_{dc}$	g_1	C+
				$U_a = U_{dc}, U_b = 0, U_c = U_{dc}$	g_2	C+B-
				$U_a = 0, U_b = 0, U_c = U_{dc}$	g_3	C+A-B-
				$U_a = 0, U_b = 0, U_c = 0$	g_1	B-
011→001	CA→CB	C	$e_a = -E, e_b = -E, e_c = E$	$U_a = U_{dc}, U_b = 0, U_c = U_{dc}$	g_2	A+B-
				$U_a = 0, U_b = 0, U_c = U_{dc}$	g_3	C+A+B-
001→101	CB→AB	B	$e_a = E, e_b = -E, e_c = E$	$U_a = U_{dc}, U_b = 0, U_c = 0$	g_2	A+B-
				$U_a = U_{dc}, U_b = 0, U_c = U_{dc}$	g_3	C+A+B-


FIGURE 6. Three phase winding conduction of A+B-C-.

Arrange the (1) to get the line voltage expression:

$$\begin{cases} u_{ab} = R \cdot (i_a - i_b) + L \frac{d(i_a - i_b)}{dt} + e_a - e_b \\ u_{bc} = R \cdot (i_b - i_c) + L \frac{d(i_b - i_c)}{dt} + e_b - e_c \\ u_{ca} = R \cdot (i_c - i_a) + L \frac{d(i_a - i_a)}{dt} + e_c - e_a \end{cases} \quad (31)$$

Subtracting U_{ca} from U_{ab} to get:

$$u_{ab} - u_{ca} = R(2i_a - i_b - i_c) + L \frac{d(2i_a - i_b - i_c)}{dt} + 2e_a - e_b - e_c \quad (32)$$

If it is in the commutation process, (33) can be sorted out:

$$U_{dc} = \frac{3}{2} Ri_a + \frac{3}{2} L \frac{di_a}{dt} + 2E \quad (33)$$

At this time, the current i_a is the non-commutation phase and its magnitude is positive. where $\frac{di_a}{dt}$ is approach 0, then $U_{dc} > 2E$ is permanently valid. Therefore, the current rate

of formula (30) is positive. It can be used to suppress the decrease in current rate in formula (26).

Based on the logical sequence of the six-step commutation and the above analysis of switch states, we can perform switch state insertion compensation for the commutation phase after normal conduction as shown in Table 1.

D. SELECTION OF COST FUNCTION

To ensure a consistent current that does not alternate, it becomes essential to optimize the control parameters using the cost function in MPC. As the predicted output current $\hat{i}_p(k+1)$ might accurately signify the future actual output current, the cost function aims to minimize the disparity between the reference current $i^*(k+1)$ and the actual current. Consequently, the cost function can be chosen in the manner described below:

$$g = |i^*(k+1) - \hat{i}_p(k+1)| \quad (34)$$

Assuming there are three conduction modes for switches, each mode corresponds to a different cost function (g_1, g_2, g_3) as indicated in Table 1. The optimal switch state for insertion can be determined based on the cost function with the lowest value. This minimum cost function can be mathematically represented as:

$$g = \min \begin{cases} g_1 = |i^*(k+1) - \hat{i}_{p1}(k+1)| \\ g_2 = |i^*(k+1) - \hat{i}_{p2}(k+1)| \\ g_3 = |i^*(k+1) - \hat{i}_{p3}(k+1)| \end{cases} \quad (35)$$

E. FEEDBACK CORRECTION

As the motor operates, the system's parameters undergo alterations, leading to discrepancies between the actual system output and the predicted model output. Consequently, this

TABLE 2. BLDCM Parameters

Parameter	Value	Parameter	Value
Rated voltage (V)	24	Stator resistance (Ω)	0.415
Rated speed (rpm)	3000	Stator inductance (mH)	1.24
Rated power (W)	100	BEMF constant (V/kRPM)	4.27
Polar pairs	4	PWM frequency (kHz)	20
Rated torque (N·m)	0.32	Sampling time (s)	1×10^{-6}

approach incorporates the implementation of online feedback correction [25]. By identifying disparities between the anticipated and actual outputs, it becomes feasible to compute the subsequent error between the projected output current and the real output current in the system at k moment.

$$e(k) = \hat{i}(k) - i(k) \quad (36)$$

In order to lessen or even completely eliminate the discrepancy between the actual value and the output value, the error $e(k)$ is used for feedback correction of the prediction model. Following correction, the output is:

$$\hat{i}_p(k+1) = \hat{i}(k+1) - ce(k) \quad (37)$$

The error correction factor, denoted as c , typically ranges between 0 and 1. In instances where the parameters are well-matched, the error between the predicted output and the actual output approaches 0. Therefore, the minimum value of c can be considered as 0. Conversely, in case of parameter mismatch, the error between the predicted output and the actual output is relatively high. In such cases, the maximum value of c can be set to 1.

IV. SIMULATION RESULTS

The current and torque waveforms of the proposed technique were evaluated in MATLAB-Simulink for a model of BLDCM at three different velocities. The obtained results were compared with other approaches mentioned in the reference. The parameters of the BLDCM can be found in Table 2.

In this simulation, the added load size is $0.1 \text{ N} \cdot \text{m}$, so the calculation formula for torque ripple is:

$$T_{\text{error}} = \frac{|\bar{T}_u - 0.1| + |\bar{T}_d - 0.1|}{0.1} \times 100\% \quad (38)$$

where $\bar{T}_u$ is the average torque of all torques greater than $0.1 \text{ N} \cdot \text{m}$ within a current cycle, and $\bar{T}_d$ is the average torque of all torques less than $0.1 \text{ N} \cdot \text{m}$ within a current cycle.

The torque and current waveforms generated by various control strategies at 750 rpm are shown in Fig. 7. At low speeds, the current waveforms of the four control strategies resemble one another fairly closely and remain remarkably stable throughout the commutation procedure. The torque ripple of proposed method is comparatively small. According to the above torque ripple formula, it can be calculated that the

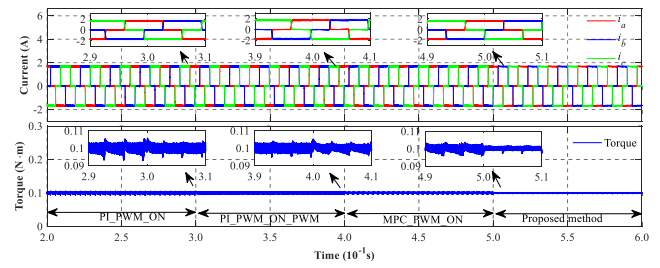


FIGURE 7. Simulation waveforms at 750 rpm.

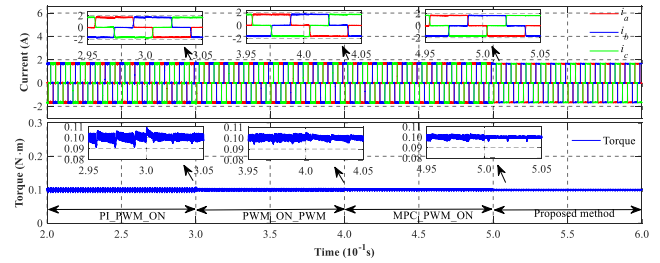


FIGURE 8. Simulation waveforms at 1500 rpm.

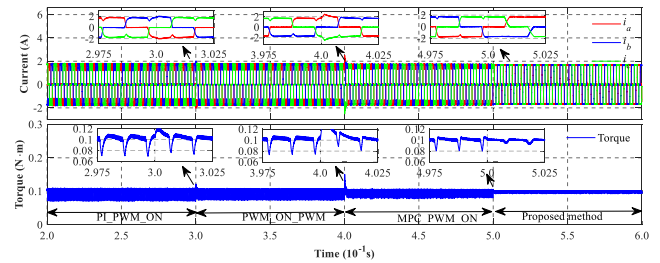


FIGURE 9. Simulation waveforms at 3000 rpm.

torque ripples under the four control schemes are 8.1%, 6.3%, 5.3%, and 2.6%, respectively.

As the speed increases to 1500 rpm, the simulation waveforms under the four control schemes are displayed in Fig. 8. It can be observed that the current waveforms change throughout the commutation process when using the PI_PWM_ON control scheme. This control scheme, which utilizes the PI_PWM_ON technique, helps to reduce current fluctuations during the commutation process, leading to a reduction in torque fluctuations. On the other hand, the MPC_PWM_ON scheme effectively maintains the non-commutation current. By implementing this suggested technique, both torque ripple and current fluctuations can be effectively suppressed. According to (38), the torque ripple for the four different schemes at 1500 r/min can be determined to be 10.7%, 8.1%, 6.4%, and 4.2%, respectively.

At a high speed of 3000 rpm, the simulation waveforms are shown in Fig. 9. It is evident that the PI_PWM_ON scheme exhibits the highest torque ripple, while the proposed scheme demonstrates the lowest torque ripple, regardless of whether we examine the current waveform or the torque waveform. The former three techniques only modify the PWM

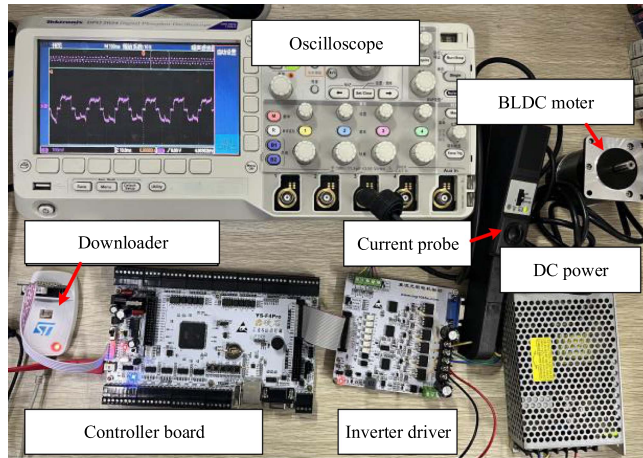


FIGURE 10. Experimental setup.

duty cycle and are ineffective in reducing the fluctuation of non-commutation current. Conversely, the proposed technique achieves a large suppression effect by keeping the non-commutation current constant, which effectively diminishes torque ripple at high speeds. The torque fluctuations are estimated to be 27.9%, 23.7%, 21%, and 8.6% for the respective control schemes.

V. EXPERIMENTAL RESULTS

The experiment platform uses the STM32F407 chip as the control chip, and the drive circuit uses MOSFET. The motor parameters match those listed in Table 2 of the simulation. Fig. 10 depicts the experimental platform. The current waveforms can be obtained through an oscilloscope, and the torque value can be computed using the electromagnetic torque equation.

A. COMPARISON OF EXPERIMENT WAVEFORMS

Fig. 11 displays the waveforms of the currents produced by various control strategies when the rotational speed is set at 750 rpm and the torque at 0.1 N · m. Both the PI control scheme and the MPC control scheme effectively regulate the current waveforms. Nevertheless, the application of the MPC control scheme yields smoother waveforms for the non-commutation currents.

When the motor speed reaches 1500 rpm and the torque measures 0.1 N · m in the middle and high-speed sections, Fig. 12 presents the current wave patterns resulting from different control approaches. Specifically, Fig. 12(a) and (b) display the waveforms of phase currents achieved under the conventional PI control strategy. It is observed that during the commutation process, the current fluctuation of non-commutation phase exceeds that measured at 750 rpm, thereby indicating a higher magnitude of torque oscillation. By implementing MPC and incorporating a duty cycle prediction technique, the torque ripple is significantly decreased, while simultaneously maintaining stability in the non-commutation current.

Fig. 13 illustrates the waveforms obtained through different control strategies, revealing the impact of motor speed 3000 rpm and torque 0.1 N · m. A noticeable result is the amplified fluctuation in non-commutation current with speed increment when the load remains constant in the traditional PI control scheme. Fig. 13(c) demonstrates the insufficiency of relying solely on MPC for preserving a steady non-commutation current through duty cycle prediction. However, employing the suggested technique proves effective in attaining a stable non-commutation current.

Fig. 14 shows the corresponding experimental torque waveforms at different speeds. At various rotational speeds, the suggested approach in this article has some inhibitory effect on torque ripple. It is consistent with simulation that there is a significant effect from suppressing torque ripple, especially at high speeds.

Fig. 15 shows the experimental current waveforms of dynamic changes in speed and load. The alterations in current waveform are evident both prior to and following the speed change, as depicted in Fig. 15(a) and Fig. 15(b). It becomes apparent that the current of non-commutation remains constant throughout the commutation stage once the speed stabilizes. As illustrated in Fig. 15(c) and (d), the current waveform of non-commutation steadily increases during the fluctuations in load. Similarly, the non-commutation current displays a tendency to maintain its level once it reaches a stable state during the commutation process.

B. PARAMETER MISMATCH AND ERROR CORRECTION

When the actual motor parameters deviate from the model parameters, it can impact the output performance of the controller [26]. In actual motor operation, the change in resistance R is relatively small and can be ignored. An increase in current can lead to magnetic saturation, which in turn leads to a decrease in inductance. That is to say, the inductance L_r in the controller are greater than the actual inductance L , so this situation is relatively close to reality. Therefore, only the influence of inductance parameter changes on the predicted current and actual output current is considered here.

To simulate the actual effect, we attempted to change the parameters of the inductance in the MPC controller, as it is difficult to change the inductance in the motor. As we reduced the inductance in the controller $L_r = 0.5L$ at 0.21 s, the predicted current i_{pre} and actual output current i_{real} Harmonic increased as shown in Fig 16(a). However, when the controller $L_r = 2L$ in Fig. 16(b), the oscillation intensifies between the predicted current and the actual output current. It can be seen that when the parameters are mismatched, the fluctuation of the predicted current will increase, leading to an increase in torque ripple as well. Therefore, MPC demonstrates a certain robustness to parameters within a certain range. Generally, the motor can maintain operation when the inductance is between $0.5L \sim 2L$. After incorporating error feedback correction at 0.22 s, the oscillation of the predicted current will be eliminated, leading to the correct switching

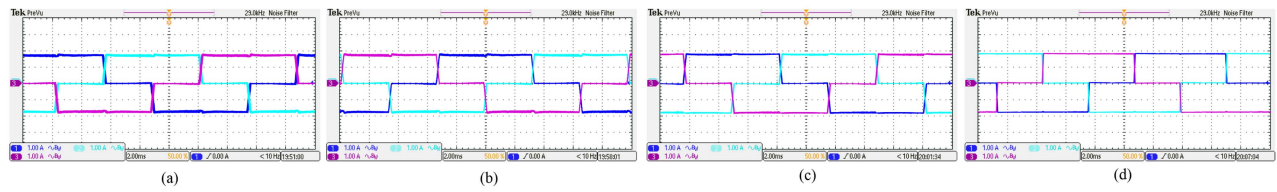


FIGURE 11. Experimental current waveforms with 0.1 N · m and 750 rpm under (a) traditional PI_PWM_ON; (b) PI_PWM_ON_PWM; (c) MPC_PWM_ON; (d) proposed method.

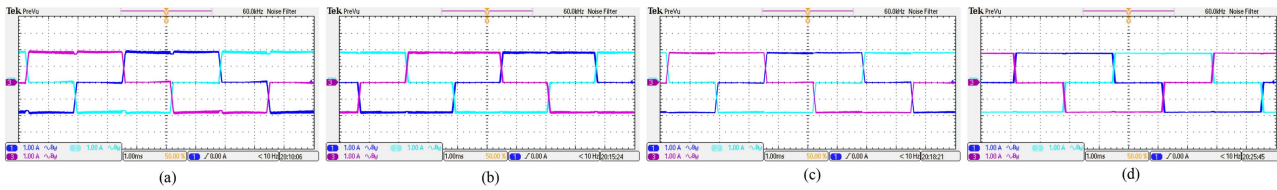


FIGURE 12. Experimental current waveforms with 0.1 N · m and 1500 rpm under (a) traditional PI_PWM_ON; (b) PI_PWM_ON_PWM; (c) MPC_PWM_ON; (d) proposed method.

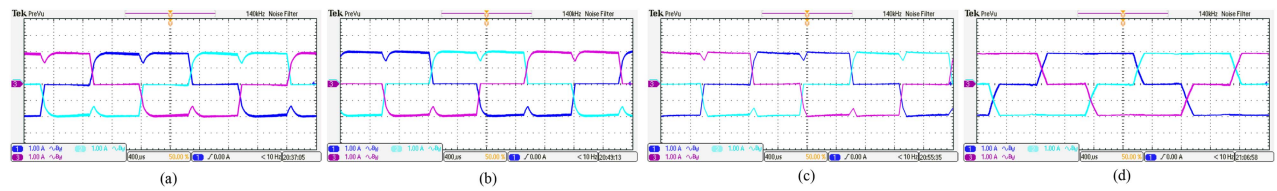


FIGURE 13. Experimental current waveforms with 0.1 N · m and 3000 rpm under (a) traditional PI_PWM_ON; (b) PI_PWM_ON_PWM; (c) MPC_PWM_ON; (d) proposed method.

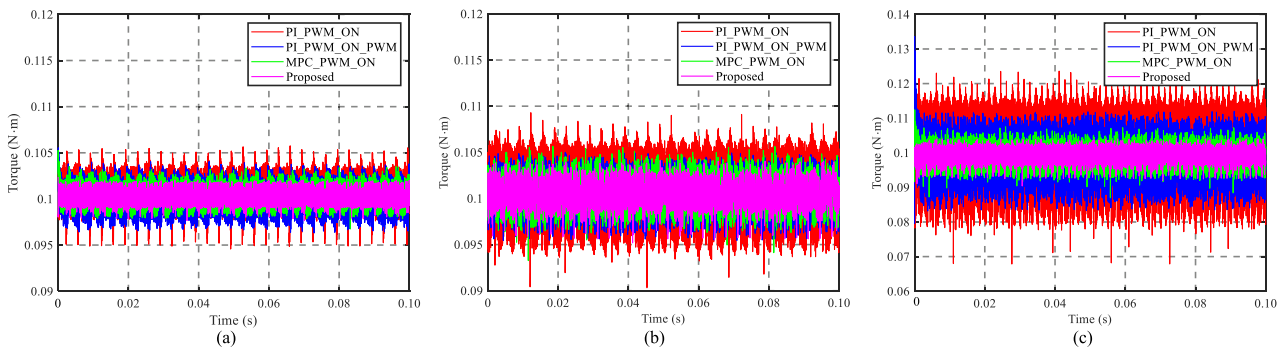


FIGURE 14. Experimental torque waveforms with different speeds at (a) 750 rpm, (b) 1500 rpm, and (c) 3000 rpm.

state output. Additionally, the actual output current fluctuation will decrease.

C. PERFORMANCE COMPARISON

In order to better compare the torque performance of the four control methods, we tested the torque ripple at different speeds as shown in Fig. 17. It can be seen that as the speed increases, the torque fluctuation also increases. Meanwhile, at the same

speed, the method proposed in this paper has the smallest torque ripple compared to other methods.

To compare its performance with other advanced algorithms, we compared its features in Table 3. These algorithms have effectively reduced torque ripple through hardware improvements, increased switching frequency, voltage compensation, and other aspects. However, under the same conditions, the proposed method can serve as a high-cost-performance ratio torque ripple suppression scheme.

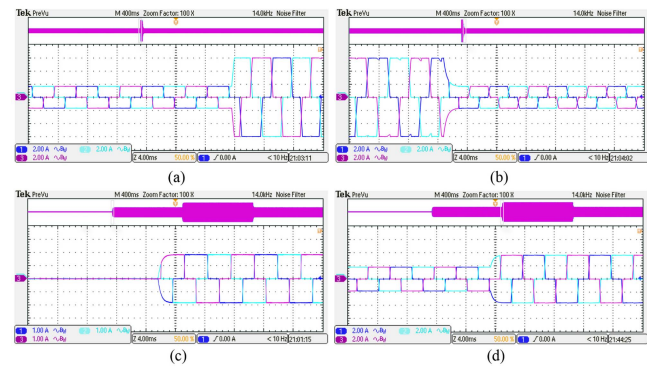


FIGURE 15. Experimental dynamic current waveforms based on the proposed method. (a) Before acceleration from 1500 rpm to 2000 rpm; (b) after acceleration from 1500 rpm to 2000 rpm; (c) load from 0 to 0.1 N · m; (d) load from 0.1 N · m to 0.2 N · m.

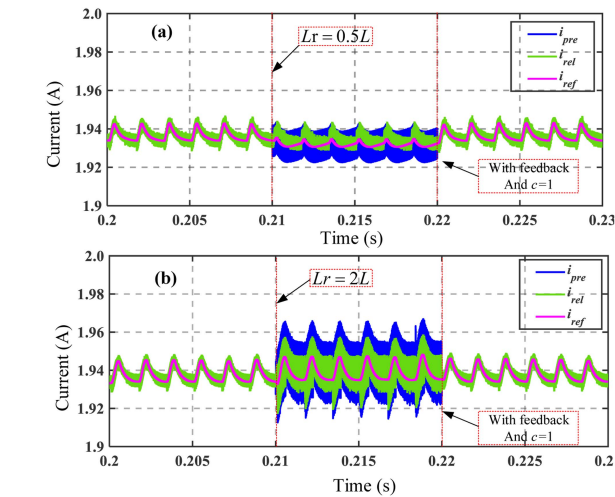


FIGURE 16. Experimental current waveforms under inductance parameter mismatch.

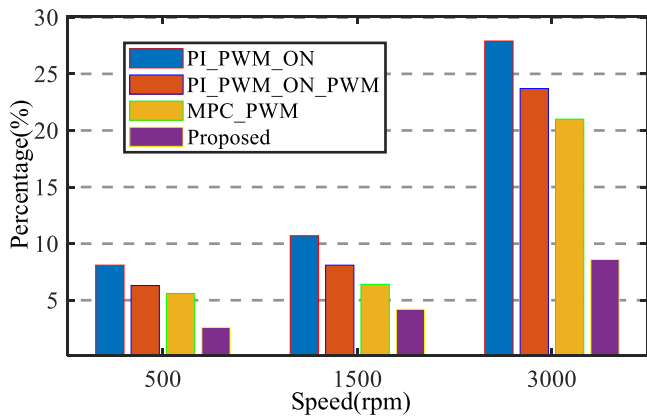


FIGURE 17. Comparison of torque ripple size at different speeds.

VI. CONCLUSION

This paper proposes an FCS-MPC strategy to address the torque pulsation problem in the non-commutation stage of BLDCM. The causes of commutation torque ripple are analyzed, and a predictive control model for non-commutation

TABLE 3. Comparison of Other Advanced Methods

Methods	Hardware Cost	Switch Frequency	Torque ripple at rated speed	Complexity	Robustness
SEPIC[3]	3-Level VSI	80kHz	6.1%	High	Low
Step-Up[4]	2-Level VSI	20kHz	12.8%	Low	High
Compensation Voltage[21]	2-Level VSI	20kHz	23%	Low	Low
Proposed	2-Level VSI	20kHz	8.6%	Middle	High

current is established. By using model prediction duty cycle instead of a PI controller, the need for PI parameter tuning is eliminated. The use of PWM_ON_PWM modulation can reduce current ripple and save hardware costs. Additionally, the insertion compensation of three switch states can minimize non-commutation current fluctuations during high-speed commutation. To account for changes in motor parameters during operation, feedback correction is introduced to reduce the error between the predicted model output and the actual output. The effectiveness of the proposed strategy in reducing commutation torque ripple at different motor speeds is verified through simulation and experimental verification. However, this method has certain requirements on the current sampling time and sampling accuracy; otherwise, it is not possible to obtain a good control effect.

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